

PANDA Component Description: *input_port*

Overview

The `input_port` component implements a simple parallel input port with adjustable width. The port can be configured to synchronize the input data, which is used for asynchronous input signals, or not over a generic.

Generics

Name	Type	Default	Description
<code>port_width</code>	integer	8	Port width (min 1, max 8)
<code>synchronize</code>	boolean	true	true → Input is synchronized false → Input is not synchronized

External Ports

Name	Type	Width	Description
<code>pin</code>	<code>std_logic_vector</code>	<code>port_width</code>	Input Port

Registers

Input Register

Name: PIN
Address: Base + 0x00

Name	PIN
R/W	R
Reset Value	-

The value of *PIN* does reflect the value of the port. *PIN* has the same width as the port (*port_width*) and is right adjusted. Unused higher order bits are always read as '0'. Write operations to *PIN* do not have any effect.